

HYDROGENATED NBR: PATENT LANDSCAPE FOR SYNTHESIS, HYDROGENATION, AND PROCESS INTEGRATION

1. ABSTRACT

The patent set spans the evolution of nitrile butadiene rubber (NBR) synthesis and its conversion to hydrogenated nitrile rubber (HNBR), with emphasis on controlling residual unsaturation, molecular weight, and downstream processability. The earliest primary patent, US3700637A, defines a highly alternating diene–nitrile copolymer architecture obtained by maintaining a nitrile-to-total monomer ratio of 0.75 to 0.92, low polymerization temperatures of 0–25 °C, and maximum conversions no greater than 35%; hydrogenated derivatives are explicitly contemplated. Later patents shift toward industrially practical NBR manufacture and post-treatment. EP2611837B1 discloses solution free-radical polymerization of at least one conjugated diene and at least one α,β -unsaturated nitrile in an organic solvent, using molecular weight modifiers such as mercaptans, mercapto alcohols, disulfides, aldehydes, halohydrocarbons, and saccharin, followed by hydrogenation. US7923518B2 focuses on NBR product stability and work-up, including calcium and chlorine contents, thio end groups, and coagulation/work-up routes that influence storage stability. US20030065076A1 and US11131058B2 are application-oriented: the former blends HNBR with hydrogenated carboxylated NBR, fillers, multivalent salts, and peroxide curing systems for abrasion-resistant track pads; the latter uses carboxy-modified NBR or HNBR latex in adhesive treatments for power transmission belt cords. The remaining patents are not NBR synthesis patents but provide context for hydrogenation and rubber compounding: US10927052B2 and US11001773B1 show selective hydrogenation strategies in hydrocarbon streams, while US10815320B2 and US11262338B2 illustrate resin modification approaches used in rubber formulations. Across the NBR-focused patents, the dominant synthesis themes are free-radical copolymerization, control of monomer feed ratio and conversion, use of chain-transfer or molecular-weight modifiers, and hydrogenation to reduce residual double bonds. The evidence supports a process window favoring controlled copolymer composition, low-temperature polymerization, and subsequent selective hydrogenation to achieve HNBR with improved stability and performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2611837B1
- Patent Title: Method for producing nitrile rubbers in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: US7923518B2
- Patent Title: Nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 3

- Patent Number: US3700637A
- Patent Title: Diene-nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1972
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 4

- Patent Number: US20030065076A1
- Patent Title: Elastomeric compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2003
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 5

- Patent Number: US11142708B2
- Patent Title: Processes and systems for petrochemical production integrating deep ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 6

- Patent Number: US10815320B2
- Patent Title: Functionalized resin having a polar linker
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: US10927052B2
- Patent Title: Decene oligomers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 8

- Patent Number: US11262338B2
- Patent Title: Modified resins and uses thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 9

- Patent Number: US11001773B1
- Patent Title: System and process for steam cracking and PFO treatment integrating selective ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 10

- Patent Number: US11131058B2
- Patent Title: Method for manufacturing core wire for transmission belt, treatment agent, and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US3700637A establishes the most compositionally constrained route: nitrile-to-total monomer ratio 0.75-0.92, polymerization at 0-25 °C, and maximum conversion no greater than 35%; hydrogenated derivatives are contemplated but no specific hydrogenation conditions are given.
- EP2611837B1 moves to solution free-radical polymerization in an organic solvent and explicitly requires a molecular weight modifier; the modifier classes are broad and include mercaptans, mercapto alcohols, mercaptocarboxylic acids/esters, thiocarboxylic acids, disulfides/polysulfides, thiourea, allyl compounds, aldehydes, halohydrocarbons, and saccharin.
- EP2611837B1 is also distinct in that it specifies a polar organic solvent with Hildebrand solubility parameter δ between 15.5 and 26 (MPa)^{1/2} and states that polymerization is followed by hydrogenation, indicating an integrated solution-polymerization/HNBR route rather than latex isolation.
- US7923518B2 does not emphasize monomer-feed constraints; instead it focuses on NBR product attributes after polymerization, including calcium content of 150-1290 ppm, chlorine content of at least 40 ppm, and specific thio end groups, with storage stability not more than 3 Mooney units.
- US7923518B2 also highlights coagulation/work-up as a key differentiator, contrasting calcium-salt coagulation, acid coagulation, and cationic-polymer coagulation approaches as routes that affect residual salts/emulsifier retention and storage stability.
- US20030065076A1 is formulation-centric rather than synthesis-centric: it combines HXNBR and HNBR with filler, multivalent organic-acid salts, and vulcanizing agent; the HXNBR is specified at 18-45 wt % nitrile, 0.3-9 wt % acid monomer, and RDB not greater than 9.0%.
- US20030065076A1 also gives an HNBR benchmark with acrylonitrile contents of 34%, 39%, and 43% and RDB values of 18, 5.5, 3.5, and less than 0.9, showing that lower residual double bond content is treated as a performance lever in the composition.
- US11131058B2 uses carboxy-modified acrylonitrile-butadiene copolymer latex or carboxy-modified hydrogenated acrylonitrile-butadiene copolymer latex in a cord-treatment

system; the patent is not about polymer synthesis but demonstrates how HNBR-derived latexes are used in adhesion formulations.

- US11131058B2 differs from the synthesis patents by emphasizing aqueous treatment agents, resorcin-formaldehyde condensate, and polycarbodiimide curing agent, with the carboxy-modified latex serving as a rubber component for fiber adhesion.
- US10927052B2 and US11001773B1 both involve hydrogenation, but in hydrocarbon processing rather than NBR chemistry; they are relevant only as process analogs for selective hydrogenation and do not disclose NBR monomer ratios, emulsifiers, or chain-transfer agents.
- US10815320B2 and US11262338B2 are resin-additive patents for rubber compositions; they support the broader application landscape for HNBR-containing compounds but do not contribute direct NBR synthesis parameters.
- Historically, the older patent US3700637A relies on tight monomer-ratio control and low conversion to obtain alternating structure, whereas the newer EP2611837B1 emphasizes solvent selection and molecular-weight regulation to enable practical solution polymerization and downstream hydrogenation.
- Across the primary NBR patents, mercaptan-type chain transfer, low residual unsaturation after hydrogenation, and careful work-up/coagulation recur as the main levers for balancing processability and final HNBR performance.

4. CONCLUSION

The evidence indicates that successful hydrogenated NBR synthesis depends on controlling copolymer microstructure before hydrogenation and minimizing residual unsaturation after hydrogenation. The strongest synthesis guidance comes from US3700637A and EP2611837B1: maintain a nitrile-rich feed window, use low polymerization temperature, limit conversion when targeting highly alternating structures, and in solution routes apply a molecular-weight modifier together with a polar organic solvent. For practical HNBR production, the patents support hydrogenating a well-controlled NBR precursor rather than attempting to compensate for poor polymer architecture later. Downstream patents show that low RDB values, controlled ionic residues, and suitable latex/coagulation or adhesive formulations are important for performance. Based on the evidence, the recommended synthesis strategy is: prepare NBR with controlled monomer ratio and molecular weight, use chain-transfer regulation where needed, then hydrogenate selectively to reduce residual double bonds while preserving nitrile functionality. For application compounds, HNBR with low residual unsaturation and compatible filler/cure systems appears most robust.

5. REFERENCES

- EP2611837B1 | Method for producing nitrile rubbers in organic solvents | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2611837B1/en>

- US7923518B2 | Nitrile rubbers | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7923518B2/en>
- US3700637A | Diene-nitrile rubbers | Not disclosed | US | 1972 | <https://patents.google.com/patent/US3700637A/en>
- US20030065076A1 | Elastomeric compositions | Not disclosed | US | 2003 | <https://patents.google.com/patent/US20030065076A1/en>
- US11142708B2 | Processes and systems for petrochemical production integrating deep ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11142708B2/en>
- US10815320B2 | Functionalized resin having a polar linker | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10815320B2/en>
- US10927052B2 | Decene oligomers | Not disclosed | US | 2021 | <https://patents.google.com/patent/US10927052B2/en>
- US11262338B2 | Modified resins and uses thereof | Not disclosed | US | 2022 | <https://patents.google.com/patent/US11262338B2/en>
- US11001773B1 | System and process for steam cracking and PFO treatment integrating selective ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11001773B1/en>
- US11131058B2 | Method for manufacturing core wire for transmission belt, treatment agent, and ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11131058B2/en>